

Exercise 1: Configuring a Basic Spring Application

pom.xml: (Updating dependencies section inside this file)

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.29</version>
  </dependency>

  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>

<properties>
  <maven.compiler.source>11</maven.compiler.source>
  <maven.compiler.target>11</maven.compiler.target>
</properties>
```

BookRepository.java:

```
package com.library.repository;

public class BookRepository {
  public void findAll() {
    System.out.println("Fetching all books from the database...");
  }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void listBooks() {
        System.out.println("Service layer: Listing books...");
        bookRepository.findAll();
    }
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.旺旺.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
                           http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository" class="com.library.repository.BookRepository" />
    <bean id="bookService" class="com.library.service.BookService">
        <property name="bookRepository" ref="bookRepository" />
    </bean>

</beans>
```

App.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class App {
    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = (BookService) context.getBean("bookService");
        bookService.listBooks();
    }
}
```

Output:

```
● adityahargunani@Adityas-MacBook librarymanagement % mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.1.0:clean (default-clean) @ librarymanagement ---
[INFO] Deleting /Users/adityahargunani/Downloads/Exercise 1- Configuring a Basic Spring Application/librarymanagement/target
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ librarymanagement ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ librarymanagement ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 4 source files to /Users/adityahargunani/Downloads/Exercise 1- Configuring a Basic Spring Application/librarymanagement/target/classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.587 s
[INFO] Finished at: 2026-06-24T17:37:05:30
[INFO]
● adityahargunani@Adityas-MacBook librarymanagement % mvn exec:java -Dexec.mainClass="com.library.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ librarymanagement ---
[INFO] Service layer: Listing books...
[INFO] Fetching all books from the database...
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.329 s
[INFO] Finished at: 2026-06-24T17:37:11:05:30
[INFO]
○ adityahargunani@Adityas-MacBook librarymanagement %
```